

MAM2000W
2LA

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Chapter 1

Eigenvectors, Eigenvalues and Diagonalisation

1.1 Eigenvectors and Eigenvalues

Reminder:

Definition 1.1.1

If A is an $n \times n$ matrix, then a non-zero vector $\mathbf{x}$ is called an eigenvector of A if $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ . The scalar is called an eigenvalue of A and $\mathbf{x}$ is said to be an eigenvector of A corresponding to the eigenvalue λ .

Proposition 1.1.1

Let A be a square matrix and $\mathbf{x}$ an eigenvector of A corresponding to eigenvalue λ . Then for every non-zero scalar μ , $\mu\mathbf{x}$ is an eigenvector of A corresponding to the eigenvalue of λ .

Proof: We have that $A\mathbf{x} = \lambda\mathbf{x}$. Now $A(\mu\mathbf{x}) = \mu(A\mathbf{x}) = \mu(\lambda\mathbf{x}) = \lambda(\mu\mathbf{x})$ ☺

Note:-

There are infinitely many eigenvectors corresponding to the same eigenvalue

Theorem 1.1.1

$$\lambda \text{ is an eigenvalue of } A \iff \det(\lambda I - A) = 0$$

Note:-

$\det(A - \lambda I) = 0 \iff \det(\lambda I - A) = 0$ so either can be used

Definition 1.1.2

Let A be an $n \times n$ matrix. The determinant $\det(\lambda I - A)$ is a polynomial of degree n in λ and is called the characteristic polynomial of A . The equation $\det(\lambda I - A) = 0$ is called the characteristic equation of A .

Note:-

It's not hard to see that if A is an $n \times n$ matrix, then the characteristic polynomial of A has degree n and the coefficient of λ^n is 1. We also know that since the characteristic polynomial of A has degree n , then the characteristic equation of A has at most n distinct roots, so A has at most n distinct eigenvalues. Since there are polynomial equations with no real roots, a matrix may have no *real* eigenvalues. It is therefore easy to find real matrices which have no real eigenvalues (see Example 4.1.8). The fundamental theorem of algebra guarantees the existence of n complex roots (not necessarily distinct, of course) for every polynomial equation of degree n . It follows that if we allow complex eigenvalues, every matrix (real or complex-valued) has eigenvalues.

Example 1.1.1

Example 4.1.7 We find all the eigenvalues of the matrix

$$A = \begin{pmatrix} 5 & -7 & 7 \\ 4 & -3 & 4 \\ 4 & -1 & 2 \end{pmatrix}.$$

As a first step, we find

$$\lambda I - A = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} - \begin{pmatrix} 5 & -7 & 7 \\ 4 & -3 & 4 \\ 4 & -1 & 2 \end{pmatrix} = \begin{pmatrix} \lambda - 5 & 7 & -7 \\ -4 & \lambda + 3 & -4 \\ -4 & 1 & \lambda - 2 \end{pmatrix}.$$

The characteristic polynomial is then

$$\begin{aligned} |\lambda I - A| &= \begin{vmatrix} \lambda - 5 & 7 & -7 \\ -4 & \lambda + 3 & -4 \\ -4 & 1 & \lambda - 2 \end{vmatrix} \\ &= (\lambda - 5) \begin{vmatrix} \lambda + 3 & -4 \\ 1 & \lambda - 2 \end{vmatrix} - 7 \begin{vmatrix} -4 & -4 \\ -4 & \lambda - 2 \end{vmatrix} - 7 \begin{vmatrix} -4 & \lambda + 3 \\ -4 & 1 \end{vmatrix} \\ &= (\lambda - 5)(\lambda^2 + \lambda - 2) - 7(-4\lambda - 8) - 7(4\lambda + 8) \\ &= \lambda^3 - 4\lambda^2 - 7\lambda + 10 \end{aligned}$$

Example 1.1.2

Thus the characteristic equation is

$$\lambda^3 - 4\lambda^2 - 7\lambda + 10 = 0$$

Having reached this stage, we are faced with solving a cubic equation. Fortunately, this one factorizes as

$$(\lambda - 1)(\lambda - 5)(\lambda + 2) = 0$$

which means that the eigenvalues are 1, 5 and -2 .

Example 1.1.3

Example 4.1.8 Let

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

The characteristic equation of A is $(1 - \lambda)^2 + 1 = \lambda^2 - 2\lambda + 2 = 0$. This equation has no *real* roots, and therefore A has no real eigenvalues. The equation has two complex roots:

$$\lambda = \frac{-(-2) + \sqrt{(-2)^2 - 4 \times 1 \times 2}}{2} = 1 + i \text{ and } \lambda = \frac{2 - \sqrt{-4}}{2} = 1 - i.$$

Therefore A does have two complex eigenvalues.

Theorem 1.1.2

Let the eigenvectors of the $n \times n$ matrix of A be $\lambda_1, \lambda_2, \dots, \lambda_n$, then

- $\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii}$
- $\lambda_1 \lambda_2 \cdots \lambda_n = \det(A)$

Proof: We'll ask you to check these results for 2×2 and 3×3 matrices in the exercises. The proof for the general case is similar, but needs properties of determinants we have not covered ☺

Note:-

Remember $\sum_{i=1}^n a_{ii}$ is known as the trace of the matrix A and denoted by $\text{tr}(A)$

Theorem 1.1.3

Let λ be an eigenvalue of the $n \times n$ matrix A . The eigenvectors of A associated with λ are the non-zero vectors in $NS(\lambda I - A)$ and hence together with the zero vector, form a subspace of $\mathbb{R}^n$ (or $\mathbb{C}^n$ if A is complex)

Proof: TODO:

An important fact that we see almost immediately is that the eigenvectors associated with an eigenvalue "almost" form a vector subspace. To see this, suppose that λ is an eigenvalue of A . A non-zero eigenvector x , associated with λ , is a solution of the equation $(\lambda I - A)x = 0$. This means that it is a member of the nullspace of $\lambda I - A$.

Moreover, all members of this nullspace, except the zero vector, are eigenvectors corresponding to the eigenvalue λ $\odot$

Definition 1.1.3

Let A and λ be as in the above theorem. Then $NS(\lambda I - A)$ is called the eigenspace associated with λ . We use notation W_λ for this eigenspace

Example 1.1.4

Example 4.1.14 We find bases for the eigenspaces of

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}.$$

The usual calculations (do them!) give the characteristic equation as

$$\lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 1)(\lambda - 2)^2 = 0.$$

Thus the eigenvalues are 1 and 2. (We say that $\lambda = 2$ is a **repeated eigenvalue** because $(\lambda - 2)^2$ is a factor of the characteristic polynomial.)

To find W_1 , we find the nullspace of $I - A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & -1 & -1 \\ -1 & 0 & -2 \end{pmatrix}$. We reduce $I - A$

to its row-echelon form $U = \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. Now U has rank 2 and nullity 1, so

our eigenspace W_1 will also have dimension 1. Suppose $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ is a member of the eigenspace; then it is a solution of $U\mathbf{x} = \mathbf{0}$. From the form of U we can deduce that x_3 is a free variable; let $x_3 = t$, where $t \in \mathbf{R}$; then $x_2 = t$ and $x_1 = -2t$, so

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = t \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}.$$

Thus $\left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$ is a basis for W_1 .

W_2 is the nullspace of $\begin{pmatrix} 2 & 0 & 2 \\ -1 & 0 & -1 \\ -1 & 0 & -1 \end{pmatrix}$. This matrix has rank 1, so must have nullity

2. Any basis for W_2 must therefore have two elements. We reduce the matrix to its row-echelon form $U = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. If $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ is a solution of $U\mathbf{x} = \mathbf{0}$, then both x_2 and x_3 are free variables. Let $x_3 = t$ and $x_2 = s$ where $s, t \in \mathbf{R}$. Then $x_1 = -x_3 = -t$, so

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = s \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

Then a basis for W_2 is $\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$

Example 1.1.5

Example 4.1.15 We have seen that the matrix $A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ has the complex eigenvalues $1 + i$ and $1 - i$. The eigenspace corresponding to the eigenvalue $1 + i$ is the nullspace of the matrix

$$\begin{pmatrix} 1+i-1 & 1 \\ -1 & 1+i-1 \end{pmatrix} = \begin{pmatrix} i & 1 \\ -1 & i \end{pmatrix}.$$

The second row of this matrix is i times the first row, so that we can reduce it to the matrix $U = \begin{pmatrix} i & 1 \\ 0 & 0 \end{pmatrix}$. If the vector $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ is in the nullspace of U , then $ix_1 + x_2 = 0$, or $-x_1 + ix_2 = 0$. Hence $x_1 = ix_2$. The eigenspace corresponding to the eigenvalue $1 + i$ is therefore

$$W_{1+i} = \left\{ \lambda \begin{pmatrix} i \\ 1 \end{pmatrix} : \lambda \in \mathbf{C} \right\}.$$

Note that in this case we have complex eigenvectors.

The eigenspace W_{1-i} can be found in the same way.

We finish off with a result which extends our list of conditions that are equivalent to the invertibility of a square matrix.

Theorem 1.1.4

An $n \times n$ matrix A is invertible if and only if $\lambda = 0$ is not an eigenvalue of A

Proof: If $\lambda = 0$ is an eigenvalue, then $0 = \det(\lambda I - A) = \det(-A) = (-1)^n \det(A) \implies \det(A) = 0 \implies A$ is not invertible. The converse is similar ☺

1.2 Diagonalization

Theorem 1.2.1

Let A be an $n \times n$ matrix

- If A has n linearly independent eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and P is the matrix whose j th column is $\mathbf{v}_j$, then $P^{-1}AP$ is a diagonal matrix D and the entries d_{jj} (on the main diagonal of D) are exactly the eigenvalues corresponding to the eigenvectors $\mathbf{v}_j$
- If there is an invertible matrix P such that $P^{-1}AP$ is a diagonal matrix $D = (d_{ij})$, then for each j , the j th column of P is an eigenvector of A corresponding to the eigenvalue D_{jj} and hence A has n linearly independent eigenvectors

: • P is square and has rank n since its columns are linearly independent. Now we calculate AP one column at a time:

$$\begin{aligned} j\text{th column of } AP &= A(j\text{th column of } P) \\ &= A\mathbf{v}_j \\ &= \lambda_j \mathbf{v}_j \text{ (Dfn for eigenvectors and values)} \end{aligned}$$

Now we calculate PD once column at a time

j th column of $PD = P(j$ th column of $D)$

$$= P \begin{pmatrix} 0 \\ \vdots \\ \lambda_j \\ \vdots \\ 0 \end{pmatrix} \\ = \lambda_j \mathbf{v}_j$$

So $AP = PD$ and thus $P^{-1}AP = D$

Note:-

We know P^{-1} exists because $\text{Rank}(P) = n$
Also we have that $A = PDP^{-1}$

- If $P^{-1}AP = D$, then $AP = PD$. Let $\mathbf{v}_j$ be the j th column of P . Then comparing j th columns on each side of the equation $AP = DP$, we see that $A\mathbf{v}_j = d_{jj}\mathbf{v}_j$. Hence $\mathbf{v}_j$ is an eigenvector of A corresponding to eigenvalue d_{jj} . Since P is invertible, it must have rank n and thus the columns of P must be linearly independent

☺

Example 1.2.1

Let $A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$. This has characteristic equation $(\lambda - 1)(\lambda - 2)^2 = 0$

The eigenspaces follow from this and we get:

W_2 has basis $\left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$

W_1 has basis $\left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$

The three vectors are linearly independent so we can form the matrix $P = \begin{pmatrix} -1 & 0 & -2 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$ which has as columns the 3 eigenvectors.

Now $P^{-1} = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 1 \\ -1 & 0 & -1 \end{pmatrix}$ and thus $P^{-1}AP = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Note:-

Notice that there is no preferred order for writing down the eigenvectors of A as columns of P . Once you have decided on the order though, the matrix D will have eigenvalues on the diagonal in the order that you wrote down the eigenvectors in P . This means that you can diagonalize A in different ways, depending on the order in which you do things. This is not a problem as long as you remember to write down eigenvectors and eigenvalues in the same order

Note:-

We do not need all eigenvalues to be distinct just the eigenvectors in the eigenspaces

Definition 1.2.1

We say a square matrix A is diagonalizable if there is an invertible matrix P such that $P^{-1}AP = D$ where D is a diagonal matrix. P is said to diagonalize A

Theorem 1.2.2

A is diagonalizable $\iff A$ has n linearly independent eigenvectors
 $\iff \mathbb{R}^n$ has a basis consisting of eigenvectors of A

Note:-

Our knowledge of matrix representations of linear transformations enables us to interpret this result in a very satisfactory way. Let us suppose that the $n \times n$ matrix A is diagonalizable. Then A has n linearly independent eigenvectors; let's call these $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ and the associated eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ (these need not all be distinct). The eigenvectors form a basis of $\mathbf{R}^n$; let's call it B . We can use A to define a linear transformation T from $\mathbf{R}^n$ to $\mathbf{R}^n$: put $T(\mathbf{u}) = A\mathbf{u}$ for every $\mathbf{u} \in \mathbf{R}^n$. Let's find the matrix ${}_B[T]_B$ representing T with respect to the basis B . Now we have

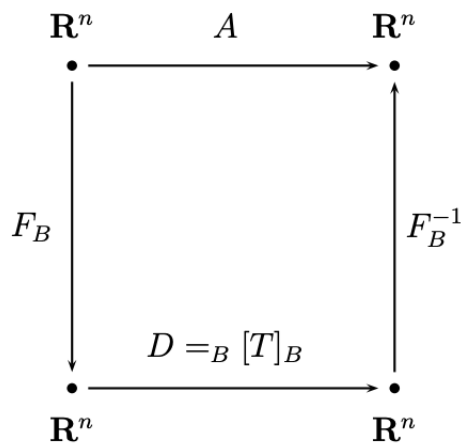
$$T(\mathbf{v}_1) = A\mathbf{v}_1 = \lambda_1\mathbf{v}_1, \quad \dots, \quad T(\mathbf{v}_n) = A\mathbf{v}_n = \lambda_n\mathbf{v}_n.$$

The coordinate vectors of the vectors $T(\mathbf{v}_i)$ with respect to B are

$$[T(\mathbf{v}_1)]_B = (\lambda_1, 0, \dots, 0), \quad \dots, \quad [T(\mathbf{v}_n)]_B = (0, 0, \dots, 0, \lambda_n).$$

These coordinate vectors are the columns of ${}_B[T]_B$. It follows that ${}_B[T]_B$ is a diagonal matrix D with the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ on the main diagonal.

If we write F_B for the linear transformation which maps a vector in $\mathbf{R}^n$ to its coordinate vector with respect to the basis B (i.e. $F_B(\mathbf{v}) = [\mathbf{v}]_B$) and recall that F_B is a bijection and therefore has an inverse F_B^{-1} , we get the following picture:



We have therefore shown that if an $n \times n$ matrix A is diagonalizable, and T is the linear transformation associated with A , then there is a basis B of $\mathbf{R}^n$ such

that the matrix ${}_B[T]_B$ representing T with respect to this basis is a diagonal matrix. The converse of this statement is also true. As a challenge, see if you can prove it. This comment also bears out the claim we made earlier, in Chapter 2, that it can sometimes be very useful to work with specially chosen bases, rather than simply the standard ones.

Warning: Not all square matrices are diagonalizable; it's a pity, but there it is!

Note:-

A sufficient condition for this is n distinct eigenvalues

Theorem 1.2.3

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m$ eigenvectors of A associated with distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$. Then $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ are linearly independent.

Proof: Suppose that S is not linearly independent ; we'll derive a contradiction. We certainly know that $\{\mathbf{v}_1\}$ is linearly independent (why?) so let k be the first integer such that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is linearly dependent. Then there are scalars $\mu_1, \dots, \mu_k$ **not all zero** such that

$$\mu_1 \mathbf{v}_1 + \mu_2 \mathbf{v}_2 + \dots + \mu_k \mathbf{v}_k = \mathbf{0}. \quad (1)$$

Now multiply both sides of the above equation (on the left) by A remembering that the $\mathbf{v}_j$'s are eigenvectors; we get

$$\mu_1 \lambda_1 \mathbf{v}_1 + \mu_2 \lambda_2 \mathbf{v}_2 + \dots + \mu_k \lambda_k \mathbf{v}_k = \mathbf{0}. \quad (2)$$

Now multiply (1) by λ_k ; we get:

$$\mu_1 \lambda_k \mathbf{v}_1 + \mu_2 \lambda_k \mathbf{v}_2 + \dots + \mu_k \lambda_k \mathbf{v}_k = \mathbf{0}. \quad (3)$$

If we now subtract (2) from (3) we get:

$$\mu_1 (\lambda_k - \lambda_1) \mathbf{v}_1 + \mu_2 (\lambda_k - \lambda_2) \mathbf{v}_2 + \dots + \mu_{k-1} (\lambda_k - \lambda_{k-1}) \mathbf{v}_{k-1} = \mathbf{0}.$$

But the vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{k-1}\}$ are linearly independent (why?) so, since none of the $\lambda_k - \lambda_i$ are zero, we must have all the μ_i zero for $1 \leq i \leq k-1$. Going back to equation (1) this leaves us with $\mu_k \mathbf{v}_k = \mathbf{0}$ and since $\mathbf{v}_k \neq \mathbf{0}$, we must have $\mu_k = 0$. But this contradicts the fact that we knew that the μ_i 's were not all zero. We have a contradiction, and the proof is finished. ■

Note:-

Given $n \times n$ matrix

1. Determine eigenvalues
If you find n distinct eigenvalues, then you can diagonalize
2. Otherwise (repeated eigenvalues), it is still possible to diagonalize the matrix

Example 1.2.2

For $A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$ gives characteristic equation $(1 - \lambda)(2 - \lambda)^2 = 0$

In this case we say 2 has algebraic multiplicity 2.

Remember as a check, sum of eigenvalues is sum of diagonal entries (double counting 2 in this case)

$$W_1 \left\{ s \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} : s \in \mathbb{R} \right\}$$

$$W_2 = \left\{ s \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} : s, t \in \mathbb{R} \right\} \text{ (we say eigenvalue 2 has geometric multiplicity 2)}$$

$$P = \begin{pmatrix} -2 & -1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \text{ (if geometric multiplicity = algebraic, fine. Geometric multiplicity must always be less than or equal to algebraic multiplicity)}$$

Example 1.2.3

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \Rightarrow \lambda = 1, \lambda = 1 \text{ (triangular matrix) with algebraic multiplicity 2 } ((1 - \lambda)(1 - \lambda) = 0)$$

$$W_1 = NS \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}. \text{ We have that Nullity} = 1 \Rightarrow \text{geometric multiplicity of 1 is 1, hence not diagonalizable}$$

Example 1.2.4

Example 4.2.6 We determine whether the two matrices

$$(a) \quad A = \begin{pmatrix} 5 & -7 & 7 \\ 4 & -3 & 4 \\ 4 & -1 & 2 \end{pmatrix} \quad (b) \quad B = \begin{pmatrix} -6 & 12 & -1 \\ -6 & 11 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

are diagonalizable.

(a) You should check that the characteristic equation is

$$(\lambda - 1)(\lambda + 2)(\lambda - 5) = 0$$

so A has 3 distinct eigenvalues. It must therefore have three linearly independent eigenvectors and so must be diagonalizable.

(b) The characteristic equation of B is

$$(\lambda - 2)^2(\lambda - 3).$$

So B has only two distinct eigenvalues. It all now depends on the dimensions of the eigenspaces. Both these eigenspaces have dimension 1, which you should check; that means that we can only get at most two linearly independent eigenvectors, so B is not diagonalizable.

1.3 Applications of Diagonalization

Markov Matrices

Recall that in Example 1.1.1 we set up a matrix which described, in terms of percentages, what happened each month to the transport preferences of UCT students commuting to campus. The matrix was

$$M = \begin{pmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{pmatrix}$$

This is an example of a stochastic, Markov, or probability matrix. We give the general definition now.

Definition 1.3.1

A square matrix $M = (m_{ij})$ is called a Markov Matrix if

- $m_{ij} \geq 0$ for all i and j
- for each j , $m_{1j} + m_{2j} + \dots + m_{kj} = 1$
(the entries in each column add up to 1)

Note:-

These are also called transition matrices

Lets look at some examples where diagonalizing a matrix is useful:

Example 1.3.1

Let's make the following assumptions about UCT students and their transport preferences:

- Ten thousand students have to get to the UCT campus every day; this number stays constant.
- At the start of each month, 20% of students who were using the Jammie Shuttle change to using their own transport and 10% of students using their own transport change to using the Jammie Shuttle.
- At the beginning of the first month of the academic year, 6000 students use the Jammie Shuttle

We want to explore the way in which the transport preferences will change with time, and what the long term trend (if any) will be

First let us denote $x(n)$ as the number of students using the Jammie at the beginning of month n

$y(n)$ is the number of students using their own transports

This means that

$$\begin{aligned} x(n+1) &= 0.8x(n) + 0.1y(n) \\ y(n+1) &= 0.9y(n) + 0.2x(n) \end{aligned}$$

So we have that:

$$\begin{pmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{pmatrix} \begin{pmatrix} x(n) \\ y(n) \end{pmatrix} = \begin{pmatrix} x(n+1) \\ y(n+1) \end{pmatrix} \text{ At the beginning of month 2:}$$

$$\begin{pmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{pmatrix} \begin{pmatrix} 6 \\ 4 \end{pmatrix} = \begin{pmatrix} 5.2 \\ 4.8 \end{pmatrix}$$

so there will be 5200 Jammie users and 4800 using their own transport

but now we can get general formula $\begin{pmatrix} x(n) \\ y(n) \end{pmatrix} = \begin{pmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{pmatrix}^n \begin{pmatrix} x(0) \\ y(0) \end{pmatrix} = \begin{pmatrix} 0.8 & 0.1 \\ 0.2 & 0.9 \end{pmatrix}^n \begin{pmatrix} 6000 \\ 4000 \end{pmatrix}$

Now if we diagonalize some matrix $A = SDS^{-1}$ (if $D = S^{-1}AS \implies A = SDS^{-1}$), then

$$A^n = (SDS^{-1})^n = \underbrace{SDS^{-1}SDS^{-1}\dots SDS^{-1}}_{n \text{ times}} = SD^nS^{-1}$$

Now for $D = \begin{pmatrix} d_{11} & 0 & \dots & 0 \\ 0 & d_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_{mm} \end{pmatrix}$, we get that

$$D^n = \begin{pmatrix} d_{11}^n & 0 & \dots & 0 \\ 0 & d_{22}^n & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_{mm}^n \end{pmatrix}$$

and we use this to calculate A^n . The steady-state vector is one where $\begin{pmatrix} x(n+1) \\ y(n+1) \end{pmatrix} = \begin{pmatrix} x(n) \\ y(n) \end{pmatrix}$, that is, we take $n \rightarrow \infty$ and get the matrix the state tends towards

Example 1.3.2

Differential equation example